

RAPORTI FINAL · FINAL REPORT

Counting the March

*Bulevardi Dëshmorët e Kombit, Tirana — 10 June 2026, 19:57–20:27 · Piramida →
Universiteti*

≈ 22,000

protesters in the boulevard corridor

locked band: **19,000 – 25,000**

A measured count of every person who marched past a hovering drone —

29 minutes 53 seconds of 4K night footage, one file: `DJI_20260610195734_0552_D.MP4`

Every person counted exactly once. Every factor in the count is a hand measurement with a saved, marked image.

Executive Summary

≈22,000 people [19,000 – 25,000] marched down Bulevardi Dëshmorët e Kombit past the drone between 19:57 and 20:27 on 10 June 2026, each counted exactly once. The number is not a model fit and not an extrapolation from a single photograph. It is a four-term decomposition, every term a measurement, locked on 2026-06-12 after the project owner's own frame-by-frame audit of the final video:

The locked decomposition: 20,202 counted in **11 turnover freezes** at full-width wide rows + **611** between-freeze continuity + **993** start bookend (already past the line at $t=0$) + **150** end bookend (still upstream when counting stopped) = **21,956** ≈ **22,000** [19,000 – 25,000]

The method in five points

- 1. Turnover freezes at the wide rows.** Eleven times across the video, the frame is frozen at a row where the **entire corridor width is in frame**, and a full-width strip — sized to the crowd's *measured turnover* (the strip's population renews every ≈3.6 minutes, timed by anchored-object transit) — is counted in full. Each marcher passes through exactly one freeze strip: nobody counted twice, nobody skipped.
- 2. Detection misses measured by hand, head by head.** At the wide rows the detector sees only a fraction of the people, so entire strips were **hand-marked head by head** and the factor *true people* ÷ *detector dots* measured per era: **S3 ×3.75, S4 ×10.3, S5 ×5.27** (S1/S2 ×1.5 estimated — lit, near rows). These factors are the report's central honesty device: every unit of multiplication traces to a marked image.
- 3. Displayed dots match counted people.** The deliverable video renders P2PNet perspective-strip dot fields — ≈3× the box-detector's yield at the far rows — so what you *see* dotted in each freeze is what was *counted*, not a sparse fraction of it.
- 4. An instrument ladder, not a single machine.** Five independent instruments bracket the answer: the near-row band engine (**16.9k — the hard floor**, smallest verifiable multipliers), the hand census with conservation (**18.9k best / 22.0k ceiling** — and *no upstream surplus anywhere*, which **excludes 30k**), the gate tracking engine (raw 12.6k/14.6k, corrected into the same zone), the wide turnover engine (**22–23.6k**), and the geometric anchor (18–27k). The lock, 22,000, sits where the strongest instruments meet.
- 5. The owner audited the machine — and cut his own number.** The project owner's frame-by-frame video audit added 2 freezes and 3 strip extensions and relocated the final counting band. When that relocated freeze absorbed the upstream mass, he cut the end bookend from **1,813 to 150** — the count's integrity centerpiece (Chapter 4).

The history of the estimate is part of the result: 12,353 → 16,912 → 20,300 → 23,619 → **22,000 locked** — every move driven by a measurement the owner's challenges forced: dense-strip net factors, measured edge shares, the departure-pass end term, pacing calibrated by circled humans, a duplication catch via transit truth, and full strips hand-marked head by head.

Scope, stated plainly: this is the **boulevard corridor only**. People arriving at the destination square by side streets never crossed the counting line and are invisible to this camera. Full event attendance is higher by an amount this footage cannot measure.

Deliverables

- **FINAL_FULL.mp4** (~29:30) — the full-length count video: turnover freezes, P2P dot fields, live tally, bookends; **FINAL_SHPEJT.mp4** (2:49) — the same at 12×.
- **RAPORTI_FINAL.pdf** — this report.
- **Gate video companion** — **GATE_FULL.mp4** + **GATE_FULL_results.json**, the independent tracking-engine cross-check.
- **docs/** — the full audit trail: 30+ experiment logs, every marked image (recall curve, pacing panels, hand-marked full strips, census tiles), the decision log with every superseded estimate and its reason.
- **real/final_number.json** — the machine-readable result.
- **This report** — Chapter 1: the method's history; Chapter 2: the measured science; Chapter 3: the sidewalk correction; Chapter 4: the locked number, the instrument ladder, and the physics; Chapter 5: the final turnover architecture and the truth-tax law.

Chapter 1 — The Project & The Method

What this chapter establishes: what we set out to do — count every person who marched down Bulevardi Dëshmorët e Kombit past a hovering drone, at night, across 29 minutes and 53 seconds of 4K video — and how the method that finally did it was built. Not the method we started with: the one left standing after every earlier approach was tested against hand-verified ground truth and either fixed or discarded. The result is a **displacement-paced freeze turnstile**: 113 auditable freeze-frames, each one a still image where every counted person is a visible dot, chained together so that everyone is counted exactly once. Its output — **16,912 [13,778–20,508]** — survived every subsequent audit as the project’s **hard floor**: the instrument with the smallest verifiable multipliers. The locked final count, **≈22,000 protesters [19,000–25,000]**, is built on the wide-row turnover architecture this floor anchors (Chapters 4–5).

1.1 The task

On the evening of 10 June 2026, a drone hovered over Bulevardi Dëshmorët e Kombit in Tirana and filmed the protest march walking toward and underneath it. One file:

Footage	DJI_20260610195734_0552_D.MP4 — 29:53, 4K portrait (2160 × 3840), night
Goal	Count everyone who passes the camera, from the head of the march until the drone leaves its station, each person exactly once
Counting window	0 → 28:55 (t = 1735 s, when the drone departs); the people still waiting upstream at that moment are counted separately from the drone’s departure pass at 29:48
Constraint	Fully automatic counting machinery — 29 minutes is far too long for hand-drawn regions — but every automatic output verified against careful human counting

The honest scope, stated up front and repeated on the end card: this is the **boulevard corridor only**. People arriving at the destination square via side streets are not visible to this camera and are not in this number.

1.2 Why this is hard

Four enemies, all measured rather than assumed:

1. **Darkness.** The road center at the counting rows drops to luminance 16–35 (out of 255) for most of the video; only the sidewalks are lit. Detector recall was measured across the brightness range: darkness alone costs ~15–20 recall points (`limit-recall-curve.md`).



Lit vs dark bands at $t=729$ — sidewalks bright, road center near the sensor floor

2. **Scale.** A person's apparent size runs from ~ 95 px boxes near the camera to 10–15 px heads in the far field. The recall curve has a measured cliff: below a ~ 45 px box (~ 15 px head) detection **collapses below 50% at every light level**, and at *any* size once the crowd is contact-packed. Far-field counting is not “harder” — it is impossible to verify, for the machine *and* for human markers (two careful marking passes on the same far-field patch disagreed by $2\times$). All far-field patch counts were therefore **archived and excluded** from the final number (`decisions.md`, 2026-06-12).
3. **A camera that only pretends to be stationary.** Across 29 minutes the drone repositions four times and drifts continuously — position, altitude (scale $\pm 20\%$), gimbal pitch, and yaw all change. Consequence, adopted as a binding rule: **nothing measured in pixels transfers between camera positions**. Every quantity — head size at a row, street width, pacing — must be re-derived per era (`decisions.md`; `05-operating-bands.md`).
4. **Occluders.** Carried banners and flags both hide the people under them (one calibration patch lost $\sim 40\%$ of its area to a single banner) and generate false detector dots on the cloth. Both effects were measured and corrected, not ignored.

1.3 The journey: three generations of method

Generation 0 — the precedent (v1, dusk, 43 seconds)

An earlier 43-second moving-drone clip of the same movement was counted with hand-drawn full-width regions: **7,548**, explicitly a floor (the detector was later measured to miss ~42% in dense dusk crowd). Two lessons carried forward: hand-counted calibration patches are mandatory, and hand-drawn regions do not scale to 29 minutes.

Generation 1 — the tracking gate (built, audited, demoted)

The obvious design for a stationary camera: track every person, tick a counter when a track crosses a line. It was built (`gate_v2`) — and then reverse-audited against hand truth, which killed it as a primary method:

- The tracker produced **3,074 track fragments with a median lifetime of 0.33 s — about 10 fragments per real person** in the dense night crowd.
- In two hand-verified 300 px windows, the gate fired **129 times for 11–17 actual crossers**: an over-count of **2–3×**, driven by track fragmentation and flag cloth bobbing inside the trigger band (`rethink-dedup`, takeover memo `03-takeover-decisions.md`).

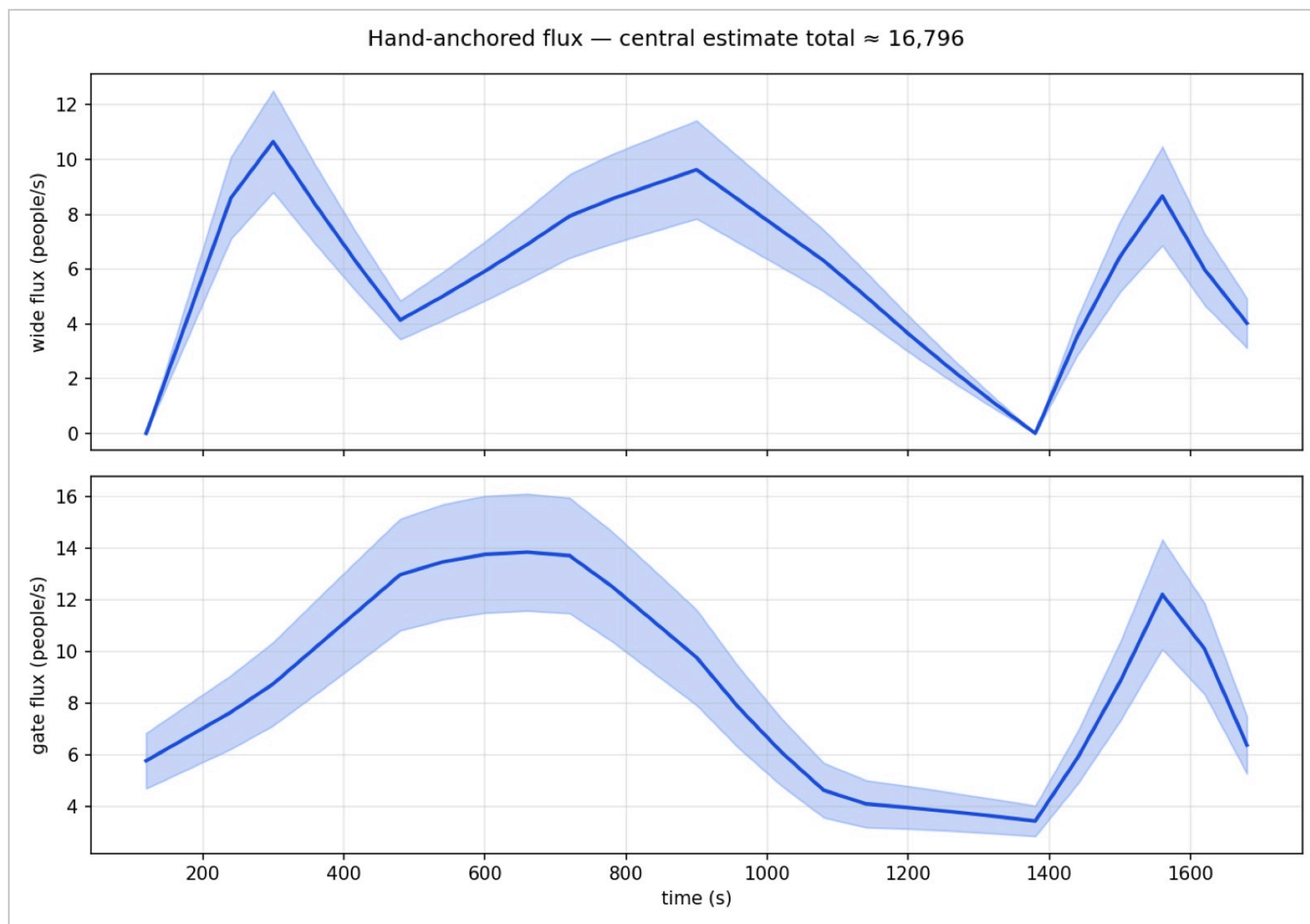
The failure mode is worth stating plainly, because it is the failure mode of most “AI crowd counting” claims: per-frame detection was fine — **identity over time is what breaks** in a dense, dark, deformable crowd. Methods were rebuilt so that *no person’s identity ever needs to be tracked* to be counted. The patched gate survives only as an independent cross-check.

Generation 2 — physics flux (right shape, soft inputs)

Next came a trackless integral — the standard line-of-interest formulation:

TOTAL = people already past the line at $t=0$ + $\int \text{density} \cdot \text{width} \cdot \text{speed} dt$ + people still upstream at the end

with density from hand-counted patches, width from a van-calibrated px/m model, and speed from optical flow. This produced $\approx 16,800$ crossings (band 13,764–19,828), agreeing with the de-duplicated gate cross-check ($\sim 16,900$), and a public working estimate of $\approx 20,000$ [17–27k] (`04-final-count.md`). It also settled an important argument with arithmetic: the end of the video shows an **empty boulevard plus escort vans** — the march tail *did* pass. A 50,000 corridor total would require ~ 28 people/s sustained for 29 minutes; the measured peak is 12–20/s. Not supportable.



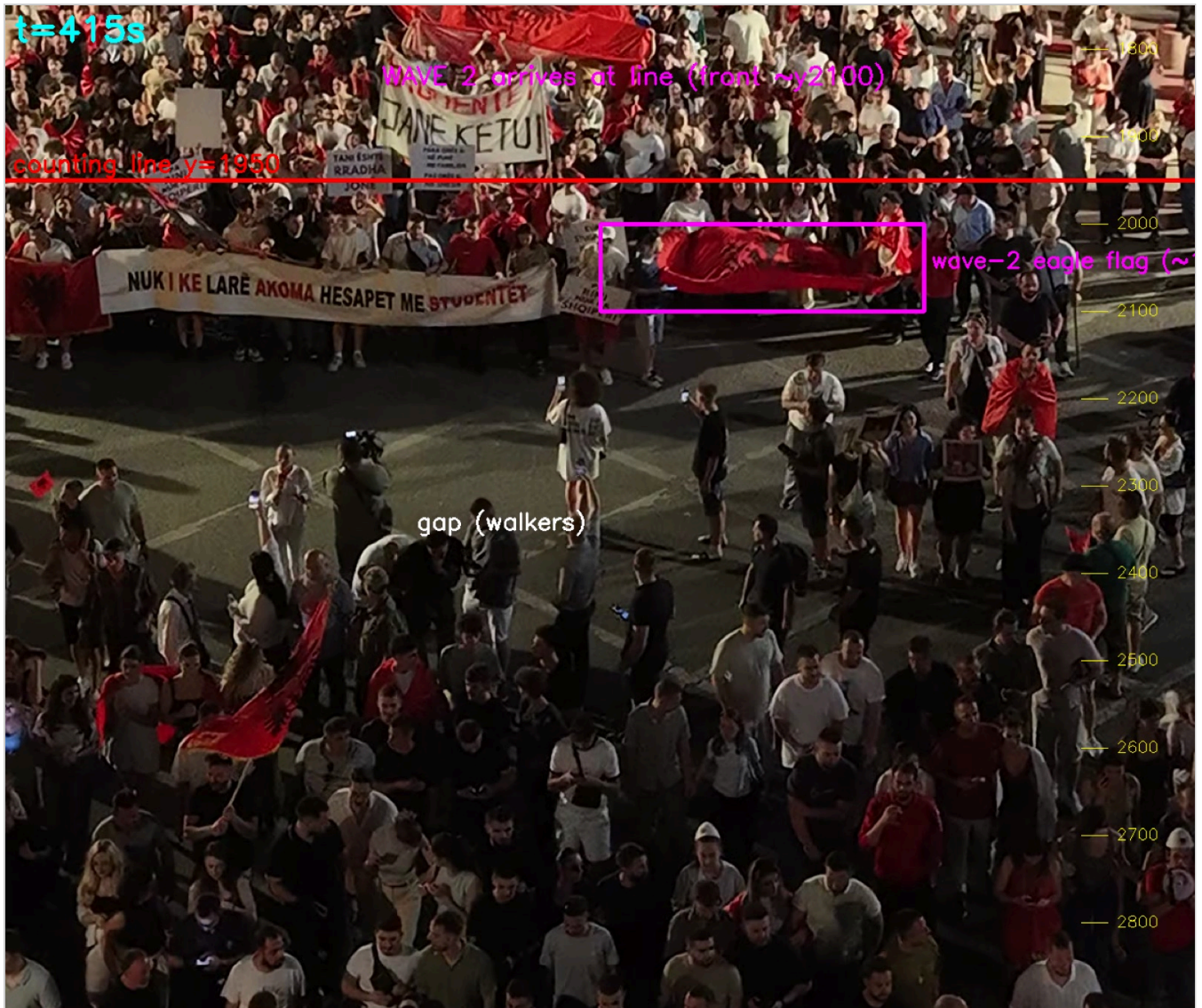
The flux integral over the full video — the area under this curve is the Generation-2 count

But two soft inputs remained, and both were exposed by direct human measurement:

Soft input 1: optical-flow pacing. A by-eye audit followed distinctive individuals (a teal shirt, a red cap, a phone flashlight...) across 35-second windows on coordinate grids and compared their true displacement to the engine's optical-flow claim. In dense flow the engine was *correct* (claimed 353 px, measured ~ 360 px — ratio 1.02). But in the lull between two waves it read **half the truth** (claimed 225 px, measured ~ 450 px; people actually crossing the line moved at 7–13 px/s against a claimed 3 px/s): optical flow locks onto the slow receding tail and misses the sparse fast walkers (`transit-audit.md`). This confirmed, with named, circled individuals, the project owner's own eyeball diagnosis that the engine was “missing 60%” in the dark eras.



Interval A of the transit audit — circled, labelled individuals tracked by eye across 35 s



Interval B — the inter-wave lull where optical flow read half the true displacement

Soft input 2: one geometry for six camera positions. The flux model initially reused pixel quantities across the whole video. The per-era re-derivation (rule 3 above) replaced this with the **operating-band table**: one counting row per camera era, each placed where the true person-box is ≈ 95 px — double the measured detection-collapse threshold — and each validated by independent agent spot-checks (measured recall 0.73 / 0.77 / 0.77 at S3 / S4 / S5, all inside their predicted ranges; 05-operating-bands.md).

Generation 2 was not wrong — its public band ($\approx 20k$ [17–27k]) contains the locked final number, and it survives in the final instrument ladder as the geometric anchor. It was *unfinished*: the pacing and the geometry needed to become measurements instead of model outputs. The final machine is what that looks like.

1.4 The final machine: a displacement-paced freeze turnstile

The core idea is simple enough to verify by hand, which is the point:

Freeze the video. Draw a 450-pixel-tall strip across the street at the counting line and dot every person inside it. Since the last freeze, the crowd has advanced a **measured** distance D . Then exactly the fraction $D/450$ of the people in the strip are *new* — they were upstream of the strip last time. Add $\text{count} \times D/450$ to the tally. Repeat every 15 seconds for the whole video. Everyone who walks down the boulevard enters the strip from the top exactly once, so everyone is counted exactly once — **with no tracking of anyone**.

Every term in that sentence is a measurement with published evidence, and every freeze is a saved still image (`freeze_final/`, 113 of them) where each counted person is a visible red dot. The count is the dots. That is the auditability standard the tracking gate could never meet.



A counting band on the real footage (S4 era, $t=1150$): full-width strip, head dots, the only thing the tally ever sees

The smoke tests — and the junction bug story

The engine was not trusted on first build. Five **smoke runs** were executed on the same 4.5-minute slice of real footage ($t = 150\text{--}430$ s), deliberately chosen because it straddles a camera repositioning. Each run broke something, and each break changed the design:

- **Smoke 1 caught the junction bug.** When the camera jumps to a new position, the counting line must move to the new era's row. The first design handled this as a single lump correction at the boundary — and produced one enormous, unverifiable negative term (the engine changelog records it bluntly: “*smoke caught junction bug: one -1118 unreliable term*”, `nf/freeze_v3.py`). One opaque number that size, subtracting more than a thousand people in one step, is exactly what this project exists to avoid. The junction logic was redesigned from scratch (J1–J4 below).
- **Smoke 2 caught sliver hallucination.** Detection on a thin leftover ribbon of strip (16 px tall) returned **149 “people”** — pure noise. Hard rule ever since: the detector never runs on a strip under 150 px; any thin swept region is instead estimated by scaling the adjacent full strip's density, recorded as its own signed, auditable line item.
- **Smoke 3 was the eye-versus-engine pacing audit** described above (ratio 1.02 in dense flow, $\sim 2\times$ low in lulls), run on the smoke-run frames.
- **Smokes 4 and 5 settled cadence and strip height** (next two paragraphs).

The cadence decision: why 15 seconds, fixed

The elegant version of the turnstile waits exactly one strip-transit between freezes ($T = \text{strip height} \div \text{measured speed}$), so that consecutive strips tile perfectly. Smokes 1–3 ran that way — and the transit audit showed why it fails: the pacing signal is *least* reliable precisely when it is slow, so the engine stretched its freeze gaps to 30, 45, even 75 seconds during lulls — the exact moments fast walkers slip through between freezes uncounted.

The fix inverts the dependency. **Freeze on a fixed 15-second clock** (a project-owner architecture decision, recorded in the engine) and let the measured displacement D set each freeze's *new-arrival share* ($\text{count} \times$

$\min(D, 450)/450$) instead of its *timing*. At typical march speeds (~ 10 px/s) the crowd advances ~ 150 px per freeze — comfortably inside the 450 px strip, so nothing can overrun the strip between freezes. A pacing error now changes a share, not a gap: it becomes a **scale factor**, and scale factors can be calibrated against people (§1.5, step 4). With transit pacing, the same error silently deleted people.

The strip-height decision: why 450 px

Early smokes used 300 px strips and thin new-arrival slices. The final engine counts a full **450 px strip** and pro-rates (another recorded owner decision). Two reasons: a single missed or doubled detection moves a pro-rated 450 px count far less than a thin slice; and a 450 px strip is a real, legible scene a human auditor can recount from the saved still — thin slices are not.

Junction conservation (the J-rules)

Era boundaries are where double-counting and people-loss hide, so they got their own machinery, born from the smoke-1 bug:

- **J1 — no counting while the camera moves.** During each repositioning the tally pauses; displacement is carried conservatively; junction logic waits until 8 s after the new era settles.
- **J2 — the coordinate handover is measured, not assumed.** The mapping from old-frame rows to new-frame rows ($y' = a \cdot y + b$) is fitted by matching the old era's two static-background anchor templates in the new era's settled frame, with documented fallbacks.
- **J3 — the line migrates in audited steps.** The counting line walks to the new era's row in ≤ 120 px steps, one per freeze; every swept sub-strip is counted as its own **signed** record (sweep downstream: subtract; upstream: add). In the final run these migration terms sum to just **+43 people net** — the -1118 lump of smoke 1, dissolved into five small, checkable entries (`freeze_final/freezes.json`).
- **J4 — crowd speed is background-corrected.** The static scenery's apparent motion (measured on a building strip) is subtracted from the crowd's, so drone drift never masquerades as marching.

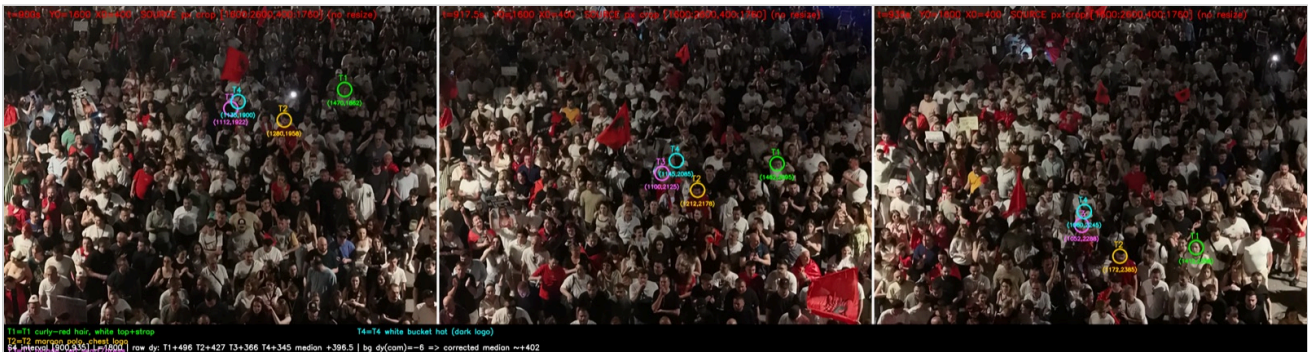
Together with the bookends (§1.5, step 8) this gives the **conservation property** the whole design is built around: *at every moment, every person on the boulevard is covered by exactly one counting mechanism* — already-counted (below the line), the turnstile strip, a signed migration strip, or a bookend. No seams, no overlaps.

1.5 The final architecture, as a pipeline

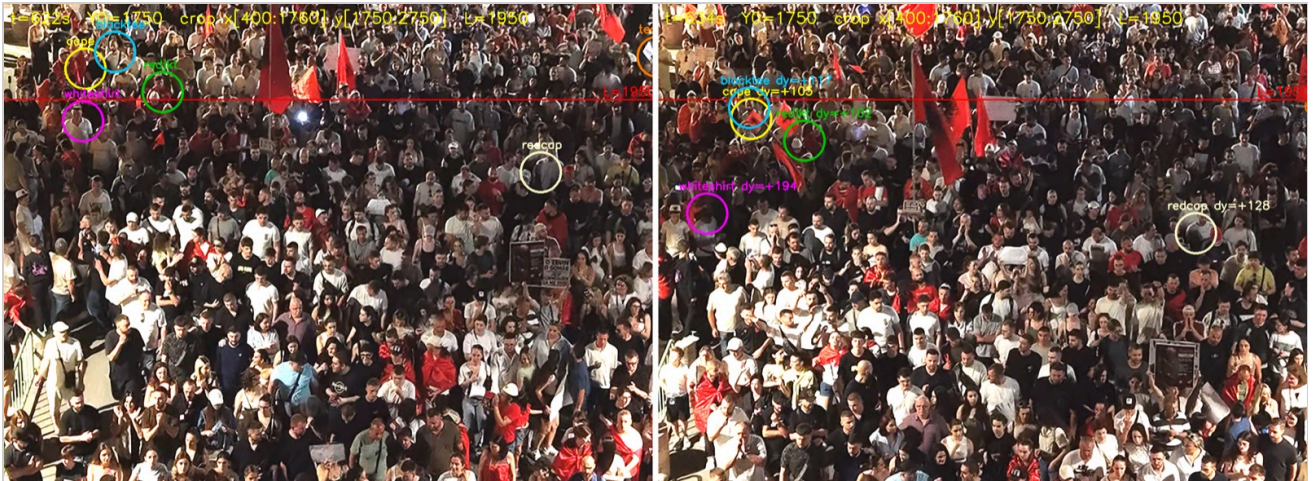
1. **Era schedule.** The video is segmented into six stable camera eras (S1, S2, S3, S4, the drift transition D, S5) separated by repositioning gaps. Each era gets a ground-anchored counting line at the row where the true person-box ≈ 95 px — twice the measured detection-collapse threshold (`05-operating-bands.md`):

era	time (s)	counting line (row)	line validated by
S1	0–135	y 2350	recall-curve interpolation, brightest cousin of S5
S2	150–240	y 1950	recall-curve interpolation
S3	255–840	y 2100	independent spot-check t=600: recall 0.73
S4	855–1320	y 1950	independent spot-check t=1150: recall 0.77
D	1320–1440	~y 1810, tracked per-frame	per-frame registration mandatory
S5	1440–1735	y 1800	independent spot-check t=1500: recall 0.77

- Ground anchoring.** Every freeze re-glues its strip to the same asphalt by template-matching static background anchors (two per era, picked on buildings); the match offset and confidence are stored in every freeze record. During era D the line is tracked frame by frame (~150 px of drift in 2 minutes).
- Pacing.** Every 5 s, crowd speed at the line is measured by forward-backward-validated optical flow on a 2 s frame pair, background-corrected (J4), and accumulated into the displacement D.
- People-calibrated pacing correction.** Because the transit audit proved optical flow under-reads in dark, stop-and-go phases, the engine's speed was calibrated **era by era against real people**: distinctive individuals were circled, re-identified by eye across 12 s and 35 s windows (with an explicit honesty log of every track that failed re-identification), and their true displacement divided by the engine's reading over the identical window. The factors: **S1 1.35 · S2 1.35 · S3 1.10 · S4 2.03 · D 1.79 · S5 2.29** ($\pm 20\%$), confirming the $\sim 2\times$ under-read in the dark eras (cal-transit.md, cal-transit-shorthop.md).



Pacing calibration in the dark S4 era: the same circled people at t=900, 917.5, 935



Short-hop calibration, S3 dark era: 12-second hops where identity is certain

5. **The freeze.** Every 15 s: grab the frame, re-anchor the line, run the detector union (VisDrone-yolo11x ∪ COCO-yolo11x, conf 0.05, head point = box top + 0.12 h, 16 px duplicate merge — the best of every model configuration benchmarked on a hand-counted patch, always on **raw** unbrightened frames) over the 450 px strip, dot everyone, and add $\text{count} \times \min(D, 450)/450$ to the tally. Reset D.
6. **Junctions** per J1–J3: pause, measured handover, signed ≤ 120 px migration records.
7. **Raw tally.** Over the full run: **113 freezes, 5 handovers, 5 migration records** → **raw turnstile tally 8,851** (freeze_final/freezes.json, every record carrying its strip, displacement, dot coordinates, velocity log, and anchor confidence).
8. **Bookends.** The window is closed by two static terms, presented as flood-dotted freeze frames. **Start: 198** — everyone already downstream of the first counting line at $t=0$ (bookends/start_dots.jpg). **End: 1,400 [1,050–1,850]** — everyone still upstream when counting stops, recounted on the drone's departure pass at 29:48, when it flies over the waiting tail (bookends/end_departure_dots.jpg). The same line that splits the bookends from the crossings guarantees no person can appear in two terms.
9. **Net detection correction, measured.** Detectors miss people at night. On 15 hand-marked strips from the *real* full-run freezes — including the six densest of the dark eras — every dot was classified true/false and every visible person counted in zoomed, brightened views. The dots are almost all true (false dots ~5–8%, concentrated on held objects); the dominant error is one-sided misses. Measured net factor (true people ÷ dots): **1.15–1.26** by density stratum (dense-patch mean 1.26, full range of patch midpoints 1.07–1.55) — recal-dense-low.md, recal-dense-high-a/b.md.



A dense S4-era calibration patch: every detector dot audited against a hand count

- l0. **Width correction, measured.** The strips see the middle ~55–57% of the corridor; the share of people walking in the off-frame edges was hand-counted era by era at full-width reference rows (2,595 people individually marked): factors $\times 1.21$ to $\times 1.60$. Chapter 3 derives this in full.



A full-width reference strip (S3, $t=600$): middle vs outer zones, the basis of the edge share

- l1. **The formula** (every term measured, none assumed):

raw freeze tally 8,851 (displacement-paced at the people-calibrated factors of step 4) $\times$ **hand-measured net factors (1.15–1.26)** $\times$ **measured edge shares ($\times 1.21$ – 1.60)**, applied era by era, + **bookends (198 start, 1,400 end) # = 16,912 [13,778 – 20,508] — the hard floor**

The band propagates the $\pm 20\%$ pacing spread, the per-era net-factor and edge-share brackets, and the end-bookend bracket end to end. In the final instrument ladder (Chapter 4) this is the **floor instrument**: built exclusively from the most verifiable near-row measurements, it cannot meaningfully overcount — and the locked final, $\approx 22,000$, must and does sit above it.

- l2. **The band-engine renders** show exactly this machine: the moving counting band, the per-freeze dots, the live tally, and the flood-dotted bookend freezes (NUMERIMI_FINAL.mp4, full length; NUMERIMI_SHPEJT.mp4, 12 $\times$). The *final* deliverable videos render the locked turnover architecture instead (FINAL_FULL.mp4, FINAL_SHPEJT.mp4 — Chapter 5).

1.6 Why believe this design — and the record of changing our minds

The estimate's own history is the strongest honesty credential we can offer, because every revision traces to a specific measurement, most of them human:

estimate	method	what killed or corrected it
7,548 (floor)	v1 hand regions, dusk clip	detector miss ~42% measured by hand count
18.5k–26.3k	raw tracking gate	reverse audit: gate over-fires 2–3× (129 fires vs 11–17 real crossers)
~14k–18k	corrected gate	superseded by physics flux with hand-anchored density
≈20k [17–27k]	physics flux $\int \rho \cdot W \cdot v$	pacing proven ~2× low in dark eras by <i>circled people</i> ; one-geometry assumption invalid across camera moves
12,353 [10.6–14.5k]	freeze turnstile, first full run	net factors and edge shares re-measured on the real freezes; end bookend recounted from the departure pass
16,912 [13,778–20,508]	freeze turnstile, recalibrated	stands as the hard floor ; superseded as headline by the wide-row turnover architecture, which sees the full corridor
20,300	first wide-row turnover integration	detection factors were still estimates; full strips then hand-marked head by head
23,619	wide turnover, hand-measured factors	duplication catch via transit truth; owner's frame-by-frame audit (+2 freezes, +3 strip extensions, relocated final band, end-term self-cut 1,813 → 150)
≈22,000 [19,000–25,000]	wide turnover, owner-locked (2026-06-12)	— stands; pinned by the census-conservation ceiling; every move in this table traces to a measurement

Three properties distinguish the final machine:

- **No tracking, anywhere.** The single failure mode that breaks crowd counting at night — identity over time — is absent by construction. Nothing needs to be followed for more than the 2-second optical-flow pair, and even that is only a pacing signal, calibrated against people.
- **Everything is a still image you can recount.** 113 freeze frames with dots, 15 net-factor patches with classified dots, 8 full-width edge strips with 2,595 marked people, 24+ circled pacing anchors per era with honesty logs of failed tracks, 2 bookend floods. The number is the sum of things you can look at. (All machine-side counts to date are agent-performed zoomed-image counts; the marked images exist precisely so anyone can audit them — `docs/verify_counts/`, `docs/experiments/assets/`.)
- **Errors were hunted where they live.** Smoke tests on a deliberately hostile window caught the junction lump and the sliver hallucination before the full run; the by-eye transit audit caught the pacing under-read; the dense-patch audits caught the night miss rate; the geometry rule caught the cross-era pixel fallacy. Each fix is in the record with the bug that caused it.

What this number is: everyone who marched down the boulevard past this camera between 0:00 and 28:55, counted once, plus the measured bookends at both ends. What it is not: the event's total attendance — side-street arrivals and people already at the destination are invisible to this camera, and this footage cannot measure them.

Chapter 2 onward unpack each link in the chain: the detector and the measured recall curve, the width/sidewalk correction (Chapter 3), the pacing calibration, the net factors, and the bookends — each with its evidence images and its open caveats.

Chapter 2 — Experiments & Findings: The Measured Science

The final number in this report is not the output of a single program. It is the product of a chain of measured quantities — detector recall, crowd speed, street geometry, camera drift — each established by its own experiment, logged in `docs/experiments/`, and each backed by images that anyone can open and audit. This chapter presents that science: what we measured, what surprised us, and how every correction factor in the final formula was earned rather than assumed.

One terminology rule applies throughout, adopted as a project-wide decision: every “hand count” below is **agent-counted** — careful, zoomed, cell-by-cell counting performed by an AI agent on brightened image tiles, with every counted person marked as a dot on a saved image. These marked images exist precisely so a human can audit them. Until spot-checked by a person, they are a working basis, not ground truth — and we say so.

2.1 How far can a detector see? The recall-vs-head-size curve

The single most important measurement in the project answers a simple question: **at what point does the detector stop seeing people?** Early on we carried an unproven claim that detection “collapses at 15 px heads.” We replaced it with a measured curve (`experiments/limit-recall-curve.md`).

Method. Nine truth patches with agent-verified head positions ($n = 21$ to 1,049 people per patch) spanning head sizes, light levels, and crowd densities, plus four new 300×200 mini-patches placed at intermediate depths. The production detector — a *union* of two YOLO11x models (VisDrone-trained + COCO-trained), confidence 0.05, native resolution, raw unenhanced input — was run on each patch and every detection was judged true/false against the verified heads.

The measured curve (recall of the production union, native scale):

Detection-box height ($\approx$ head size)	Evidence (recall @ luma, n)	Recall
36–44 px (head $\sim$ 13–16 px)	t480 wide rows .40/.25/.38 @ luma 113; t280 wide .16 @ 119 (packed, $n=1049$)	.15–.40 — COLLAPSED
45–65 px (head $\sim$ 16–23 px)	N1 .84 @ 100; N2 .74 @ 81; N3 .81 @ 34; N4 .65 @ 17	.65–.84
85–96 px (operating bands)	S3 .73 ($n=59$); S5 .77 ($n=41.5$); S4 .77 ($n=54.75$) — independent spot-checks in march flow	.73–.77
100–130 px (head $\sim$ 33–48 px)	t280g .68 @ 21; t480g .71 @ 22; t729g .81 @ 49; t1100g .63 @ 73	.63–.81

Three findings define the whole counting architecture:

1. **The size cliff is real and measured.** Below box ≈ 45 px (head ≈ 15 –16 px) recall falls under 50% **at every light level we measured — including the brightest patch in the entire set.** The old “15 px” folklore turned out to be exactly right, but now it is a measurement with brackets, not a slogan.

2. **Packing is an independent kill-switch.** The patch at $t=280$ “wide” is the brightest sample we have (luma 119) and carries 1,049 verified people standing contact-packed (head spacing ≈ 1.5 head-widths). Recall there: **0.16**. Even where matched boxes reached ~ 56 px, recall stayed at 0.16. Once a crowd becomes a contact-packed wall, box detectors stop counting it — at any head size we measured, in any light.
3. **Darkness degrades but does not cause the collapse.** Going from bright to luma < 20 costs ~ 15 – 20 recall points at intermediate sizes. Size and packing dominate; light is secondary.

Evidence: matched-overlay images `assets/limit_harvest_t480_wide.png` (the far-field collapse made visible — 517 verified people, 173 matched) and the judged mini-patches `assets/limit_N1_judged.png` ... `assets/limit_N4_judged.png`.

Consequence. All counting bands were placed at rows where boxes measure ≈ 95 px — more than $2\times$ above the size cliff, in the flattest part of the recall plateau — and the far-field “wide wall” patches were **archived and excluded from the count chain entirely**: at heads of 10–15 px, two independent marking passes disagreed by $2\times$ on the same patch, so neither agents nor a human can adjudicate dots at that resolution. No patch density below the verification limit is multiplied by any area anywhere in the final number.

2.2 Findings that contradict intuition — and were kept because they were measured

Enhancement hurts the detector

The obvious move for night footage — brighten it (CLAHE/gamma) before detection — was tested on ground-truth-annotated night drone footage (`experiments/night-eval-real-analogs.md`). It **lowered recall in every single test**: 16% $\rightarrow$ 10%, 27% $\rightarrow$ 16%, 33% $\rightarrow$ 30% on the three DroneCrowd night sequences. The detector was trained on raw imagery; enhancement destroys the statistics it relies on. Rule adopted: **the detector always eats RAW frames**; brightening exists only in the human-facing views and renders.

Native resolution beats tiling and upscaling — with one measured exception

We audited every scale trick on an agent-verified 52-person ground-truth patch (`experiments/rethink-models.md`):

Variant	Count vs GT 52	Verdict
Union, native scale	46	dots on heads, ~ 6 visible misses, no noise
$2\times$ tiled inference	44	no gain
imgsz 4352 (upscaled)	46	no gain
Triple union with RT-DETR	74–100	dots on shoulders, doubles — overcount, not recall; 60 s/frame
P2PNet (SHTechA, day-trained)	24–28	out of domain at night

On night analog footage, $2\times$ upscaling *dropped* recall from 34% to 26%. SAHI-style tiling, a standard recommendation for small objects, measured zero gain here. The exception, also measured: **$2\times$ upscaled inference rescues the dark near field** — on the gate-row patches (boxes ≥ 100 px, luma ~ 21) it lifts recall from .68–.71 to **.92–.97** — but does nothing in the far field (.26 native, .26 upscaled). Upscaling helps darkness, never smallness. (One later rehabilitation, also measured: P2PNet fails on raw night frames, but run

on **perspective-normalized strips** at the wide rows it yields $\approx 3\times$ the union's dots — the dot fields rendered in the final video, Chapter 5.)

One detector is not enough: the union

No single model survives the whole night. The VisDrone model collapses in the near field after dark (down to $\sim 45\text{--}65\%$ per frame); the COCO model holds $\sim 60\text{--}80\%$ exactly there (`experiments/real-detect-real.md`: on the night patch P4, VisDrone found 65%, COCO $\sim 80\%$; in the dusk near-field patch P1, VisDrone 31% vs COCO 58%). Across ~ 50 dot-verifications on real frames, false positives were nearly absent — zero hallucinations on flags, banners, signs, or phone flashlights — so the two models' outputs could be safely **unioned** (with a 16 px centroid merge to kill duplicates). The result: where a single detector at the dark gate row sees only $\sim 60\text{--}70\%$ of people, the two-model union with upscaled inference reaches **92–97% on agent-verified gate patches** — from the low sixties into the nineties, each step verified against marked truth.

2.3 Trusting the hardware: GPU validation before a 29-minute run

All inference ran on an Apple M3 Pro GPU (MPS backend). Before committing to multi-hour runs we verified correctness, not just speed (`experiments/speed.md`): on three dense test frames (426 / 1,088 / 1,031 detections), CPU and GPU produced **box-level identical output — every single box matched at IoU ≥ 0.5 , 100% agreement** — at $5\text{--}6\times$ the speed. Evidence: `assets/correct_f30_cpu-red_mps-green.jpg` (CPU dots red, GPU dots green; every dot renders orange because the two coincide). The same benchmark fixed the production recipe: native-width band inference at 0.47 s/frame, hardware-accelerated decode at 243 fps, making the full 29-minute, 4K, 10-bit video analyzable overnight on a laptop.

2.4 Holding a line still under a drifting camera

A “stationary” drone is not stationary. Tracked against ~ 20 skyline landmarks (lit signs, facades, lamp heads), the camera frame wandered over the 29 minutes by **296 px horizontally, 730 px vertically, 16.4% in scale, and rotated to -2.05°** , with four discrete repositioning events (at ~ 140 s, ~ 250 s — a 14.5% pull-back in under 15 s — ~ 845 s, and a 2-minute slide at 1320–1440 s) (`experiments/real-drift-real.md`, trajectory plot `assets/drift_trajectory.png`).

A counting line fixed in pixels would therefore count **a different row of the boulevard at different times** — at $t=120$ s the fixed-pixel line sits ~ 500 px away from the ground row it was defined on. The evidence image `assets/gate_t0120_full.jpg` shows both: the ground-anchored line (red) glued to the same curb and railing, the naive fixed-pixel line (yellow) drifted half a screen away.

The registration that fixes this was itself stress-tested:

- **Model selection by measurement:** a full homography fit, extrapolated from skyline anchors down to the counting row, blew up to 189 px of error and was **rejected**; a 4-DOF similarity fit holds 25–60 px at the counting row, and 8–19 px when re-projecting between nearby drone stations.
- **Chained keyframes:** after the 1320 s event the original anchors leave the frame; the transform is composed through a second reference. The composition was validated by warping the $t=1700$ frame back into $t=480$ coordinates — trees, lamps, boulevard edges and the vanishing-point building all land single-imaged (`assets/chain_blend_480_1700.jpg`).

- **Error coupling:** the residual 25–60 px is absorbed by a ≥ 100 px counting hysteresis band that people cross in 1–2 s — drift cannot double- or zero-count anyone.

Because the drone’s position, altitude, pitch and yaw all change across the video, a further project rule was adopted: **no pixel quantity transfers between segments**. Head size, street width, captured fraction and the recall limit were re-derived per drone station (six segments, S1 through S5 plus the drift interval D), each with its own registered counting band.

2.5 Pacing the count: calibration by circled human beings

The counting machine works by freezes: count everyone in a band, wait exactly until the crowd has advanced one band-height, count again. Its pacing signal was optical flow (Lucas–Kanade) on the band. The critical question: **is the engine’s speed reading true?** The user’s eyeball said no — that real displacement looked $1.5\text{--}2\times$ larger than the engine claimed.

We tested this the only honest way available: by **circling individual, recognizable human beings** — a red-caped marcher, a man in a teal shirt, a woman with a white-brim cap, a carried portrait poster — and re-finding the same person 12 or 35 seconds later, reading their displacement off the source pixel grid (`experiments/transit-audit.md`, `experiments/cal-transit.md`, `experiments/cal-transit-shorthop.md`).

The protocol was strict and its failures are logged: an interval only counts if **at least four people are confidently re-identified at both endpoints** (verified at midpoint checkpoints, with camera drift measured on skyline landmarks and subtracted); intervals with fewer survivors are marked FAILED in the logs, with every lost track and every near-miss false match documented. Evidence images show each circled person at start, middle, and end: `assets/cal_S4_900.png`, `assets/cal_S1_60.png`, `assets/shp_S3dark_610.png`, `assets/shp_D_1370.png`, and twenty more `cal_*` / `shp_*` panels.

What the circled humans revealed:

1. **In dense, well-lit flow the engine was right.** Interval $t=305\rightarrow 340$: five tracked people gave a median displacement of 360 px vs the engine’s claim of 353 px — **ratio 1.02** (`assets/transit_A_t305.png`).
2. **In lulls and dark eras it read half the truth.** Interval $t=340\rightarrow 415$: true displacement of the people actually crossing the line was ~ 450 px vs the engine’s 225 px — **ratio ~ 2.0** , locally $2.3\text{--}4.3\times$ during the sparse inter-wave gap. In the dark eras S4 and S5 the optical flow systematically locked onto slow texture and under-read true transit by $\sim 2\times$ — this is the measured confirmation of the user’s “the count is missing $\sim 60\%$ here” diagnosis, and it is the single largest correction in the project.
3. **Crowds move in waves, not at a speed.** Twelve-second medians at the same spot swung from +21 px to +119 px between adjacent hops (stop-and-go sway, S3 era); verified individuals in one hop ranged from +3 px (a man who genuinely stopped to film) to +194 px; adjacent lanes differed by $2\text{--}3\times$ (3.4 vs 8.5 px/s at $t=1620$). Any single-number “crowd speed” is fiction; only medians over ≥ 6 verified identities per hop, per era, are usable.

The result is a **people-measured pacing factor per era** — true displacement of circled individuals divided by the engine’s reading over identical windows:

Era	S1	S2	S3	S4	D	S5
Pacing factor k	1.35	1.35	1.10	2.03	1.79	2.29

($\pm 20\%$ spread, carried into the final error band. Short 12-s hops were trusted over 35-s tracks, because long tracks inflate through perspective acceleration as people approach the camera.) The raw freeze tally is multiplied by these factors — the dark-era counts roughly double, because the machine was provably freezing half as often as the crowd’s true advance required.

2.6 The marked-patch program: we audited our own numbers

Every load-bearing count in this project went through the same discipline: **put a dot on every person, save the image, and let the dots be challenged**. Three audit campaigns shaped the final number.

a) The verify_counts audit — and the finding that killed a method

Twelve 500×300 patches used in the earlier density chain were re-marked dot-by-dot at 4× zoom (docs/verify_counts/, one marked image + one log per patch; 7 of 12 runs completed). The audit cut both ways:

Patch	Re-marked count	Original bracket	Verdict
t=280 gate	37 (32–43)	36–52	inside
t=480 gate	35 (31–44)	38–53	below bracket
t=729 gate	42 (37–55)	54–75	below bracket
t=1100 gate	32 (28–38)	35–55	below bracket
t=280 wide	1,049 (950–1,140)	285–405	3.0× outside
t=480 wide	517 (490–540)	278–392	1.5× outside
t=1650 wide	372 (330–410)	112–176	2.6× outside

The gate-row counts ran consistently **low** (median 0.74 of bracket-mid) — so the gate-based flux had to come down $\sim 20\%$. The far-field “wide” counts blew **up** by 1.5–3.0× — and worse, two marking passes on the same wide patch disagreed by 2×, proving that 10–15 px heads are below the verification limit for agents *and* humans alike. Rather than pick the convenient pass, we **discarded the entire far-field density route** (Section 2.1) and rebuilt the count on verifiable rows only. An audit that can kill a method is the only kind worth running.

b) Band spot-checks and the net-factor discovery

Each operating band was independently spot-checked at a fresh timestamp with the full production pipeline (3–4 patches of 200×200, clean-then-dotted counting, every dot judged true/false): S3 recall **.73** (n=59), S5 **.77** (n=41.5), S4 **.77** (n=54.75) — all inside their predicted ranges; no band had to move (experiments/limit-band-validate.md, experiments/p1-s4-spotcheck.md, overlay assets/p1_s4_overlay_t1150.png).

These checks produced a methodological finding: the right correction is the **net factor** (agent-count ÷ detector dots), not 1/recall — because rare false positives partially cancel misses. Using 1/recall would have over-corrected the count by $\sim 25\%$.

The net factor was then **stratified by density** on 15 hand-marked strips from the real production freezes, deliberately including the six densest shoulder-to-shoulder walls of the whole video (experiments/recal-

dense-high-a.md, recal-dense-high-b.md, recal-dense-low.md; evidence tiles
assets/recal_dha_t570_center_dotted.png and siblings):

Regime	Patches	Measured net (agent/dots)
S3 peak walls (t=540/570/765)	6	1.07–1.37, pooled 1.20
S4/S5-era walls (t=1065/1080/1410)	6	1.17–1.39, mean 1.26
Low/mid density (t=870/1335/1155)	3	1.22–1.55

The production nets adopted: **1.15–1.26** for the dense regimes — every unit of that correction traceable to a marked image. The dot audits also established that detector false positives are rare (~5–8%) and live on held objects (framed photos, raised phones, signs); the dominant error is one-sided under-counting, so these nets are floors, not guesses.

Even the *hand counting itself* was audited: counting packed cells at 4.32× zoom was checked against 8.6× double-zoom probes and found to under-count by ×1.3–1.55 in contact-packed centers (assets/recal_widthpeak_probe_480_c3a.png). All packed-cell counts were probe-corrected. We did not assume our own eyes either.

c) The edge of the frame: measured, not guessed

At the counting rows the camera frame captures only the middle ~55–57% of the boulevard corridor; people on the outer strips (sidewalk lanes, tree lines) cross unseen. The unseen share was **measured by hand**, at rows higher in the frame where the full corridor is visible, by classifying every marked person as inside or outside the band-visible window (experiments/p1-edge-flow.md, recal-width-peak.md, recal-width-late.md; strips assets/p1_edge_t60_S1.png, assets/recal_widthpeak_t480_strip_window.png, assets/recal_widthlate_t1005_strip.png):

Era	Measured edge share	Factor applied	People marked for the measurement
S1 / S2	23% / 18%	×1.30 / ×1.21	3 full-width reference strips
S3	24–25% [16–34]	×1.32	543 + 607 (t=480, t=729)
S4	38–41%	×1.60	724 (t=1005)
D / S5	~40%	×1.50 / ×1.45 (conservative)	721 (t=1500)

Two measured facts make these factors honest in both directions. People **taper toward the corridor edges** — edge density is only 0.73–0.87 of the middle — so assuming uniform density across the full width (a flat +75–82%) would over-correct; that naive correction was measured to overstate by 6–14%. And the share did **not** rise with density (the jammed t=600 scene showed the same share as the free-flowing t=300 scene), so one factor per era suffices. A flat “+30% for sidewalks” would have under-corrected the dark eras and double-counted in the early ones.

2.7 What the experiments add up to

Every factor in the final formula now has a measurement behind it:

Term	Value	Established by
Raw freeze tally (113 freezes, 450 px bands at box ≈ 95 px)	8,851	recall curve $\rightarrow$ band placement (§2.1); registration (§2.4)
$\times$ people-calibrated pacing per era	1.10 – 2.29	circled-human transit calibration (§2.5)
$\times$ net detection factor	1.15 – 1.26	15 dot-marked density strips (§2.6b)
$\times$ measured edge share	$\times 1.21 - \times 1.60$	2,600+ people hand-classified at full-width rows (§2.6c)
+ bookends (already past the line at start; still waiting at end)	198 + 1,400 [1,050–1,850]	freeze-frame flood-dot counts

Result of the band-row engine: 16,912, band 13,778–20,508 — the count’s hard floor. It is not the final headline: the same marked-patch discipline, applied to full-width strips at the wide rows (hand-marked head by head, Chapter 5), calibrated the turnover architecture whose locked output is **$\approx 22,000$ [19,000–25,000]** (Chapter 4). Where a measurement said our method was wrong — enhancement, tiling, the far-field wall, the engine’s dark-era pacing — the method changed, and the marked images that forced each change are all still on disk.

Chapter 3 — The Width/Sidewalk Correction: Full Derivation

What this chapter proves: the counting bands see only the middle ~55–57% of the boulevard corridor. The people walking in the off-frame outer strips — the sidewalks and street edges — were **measured by hand, era by era**, not assumed. The resulting correction factors are **×1.21 to ×1.60** depending on the era. Every number below traces to a marked, published image.

3.1 The problem: the camera frame is narrower than the crowd

The whole count rests on per-era *counting bands*: full-width 2160×300 pixel strips placed at the one row of each camera position where heads are large enough to count reliably (true person-box ≈ 95 px — see the recall-curve chapter). That row choice has a price.

The drone looks down the boulevard at an angle. Near the bottom of the frame — where the band must sit — the corridor is so wide in pixel terms that it **overflows the frame on both sides**. The corridor model (validated against visible street edges to $\pm 2\text{--}3\%$, `limit-segment-map.md`) gives the captured fraction directly:

$$\text{captured\%} = 2160 / (2.612 \cdot (y_{\text{band}} - y_{\text{horizon}}))$$

era	band row y	captured width	source
S1 (0–135 s)	2350	56.9%	p1-edge-flow.md geometry table
S2 (150–240 s)	1950	56.4%	p1-edge-flow.md
S3 (255–840 s)	2100	57.1–57.5%	p1-edge-flow.md; independently re-derived in recal-width-peak.md
S4 (855–1320 s)	1950	55%	05-operating-bands.md
D (1320–1440 s)	~1810 tracked	~57%	05-operating-bands.md
S5 (1440–1735 s)	1800	54–56%	05-operating-bands.md

The figure ~57% was derived twice, independently: once from the per-station vanishing-point geometry (`p1-edge-flow.md`), and once from scratch on the real full-run freezes (`recal-width-peak.md`: corridor at the S3 band row spans pixel $x = -779$ to $+2979$, of which the frame holds $[0, 2160] \rightarrow 57.5\%$). The two derivations agree.

So the band, on its own, counts the middle of the street and is blind to roughly the outer 45% of the corridor’s width — the strips where the sidewalks are.

3.2 The math: from “share outside” to a multiplier

Let s be the fraction of corridor people who, at the band row, walk in the strips the frame cannot see. Then the band observes only the complement:

```
band count = (1 - s) · true corridor count
true corridor count = band count × 1 / (1 - s)
```

The entire question reduces to: **what is s, era by era?** Everything else is one line of algebra. The factor is *not* an estimate layered on an estimate — it is the reciprocal of a directly measured share:

measured share s	multiplier 1/(1-s)
18%	×1.21
23%	×1.30
24–25%	×1.32
37.5%	×1.60

One assumption is required and is stated openly: **lateral stationarity** — people do not change lanes between the row where we can measure the full width (higher in the frame) and the band row (lower in the frame). The march flow is parallel along the boulevard at every measured station, which supports this; it was not separately verified (`p1-edge-flow.md`, caveat 3).

3.3 Two tempting shortcuts, and why we rejected both

Wrong answer #1: the naive geometric correction, $1/0.55 \approx \times 1.82$

If people were spread **uniformly** across the corridor’s full width, capturing 55% of the width would mean capturing 55% of the people, and the fix would be $\times 1.82$ (+75–82%).

Measurement says no. People **taper before the corridor edges**:

- At every full-width reference row, essentially 100% of detections fall *inside* the model corridor, and the occupied extent is only **80–88% of the model corridor width** (`p1-edge-flow.md`, geometry log).
- Where edge density was hand-counted column by column, the edges held **0.73–0.87** of the middle’s density (people per pixel-column): ratio 0.73 visible / 0.87 occlusion-corrected at $t=1005$, and 0.81 at $t=1500$ (`recal-width-late.md`). Thinner — but emphatically **not empty**.

Concretely: at S3 the uniform assumption predicts an edge share of 42.5%; the hand-measured share is 24–25%. Naive $\times 1.82$ versus measured $\times 1.32$ — the shortcut would have inflated the S3 flow by $\sim 38\%$. In the early eras the uniform model over-corrects by roughly **2.5×** the true edge contribution (`p1-edge-flow.md`, bottom line). In the late eras (S4/S5) the gap is smaller — $\times 1.82$ assumed versus $\times 1.60$ – 1.71 measured, a 6–14% overstatement — but it is still an overstatement (`recal-width-late.md`).

Wrong answer #2: a flat “+30% for the sidewalks”

A single flat factor fails in both directions at once (`07-sidewalk-explainer.md`):

- It **over-corrects S2**, where the measured share is 18% (correct factor $\times 1.21$, not $\times 1.30$).
- It **under-corrects S4 and S5**, where the measured share is 38–41% (correct factor $\times 1.60$, not $\times 1.30$) — a $\sim 19\%$ undercount of those eras.
- Worst of all, adding a flat allowance **on top of** the measured per-era factors would count the same sidewalk people **twice**. The measured shares already contain everyone in the outer strips; there is nothing

left to add.

The honest correction is one measured share per era, applied once.

3.4 How the share was measured

The trick: **earlier and higher**. Although the corridor overflows the frame at band rows, there are rows *higher in the frame* (further from the camera) where the **entire corridor width fits on screen**, tree line to tree line. At those reference rows you can see everyone — middle and edges — at the same instant.

The protocol, common to all measurements:

1. Take a real frame (raw, full resolution; the production freeze stills for S3–S5).
2. Locate the corridor edges at the reference row (visible street edges / tree lines; the corridor model cross-check at $t=1005$ matched the tree-to-tree measurement **within 1%**).
3. Project the band's visible window up to the reference row (a trapezoid, because perspective narrows with height), splitting the strip into *middle* (what the band would see) and *outer* (what it cannot).
4. **Hand-mark every person** in zoomed, brightened views — $4.3\text{--}8.6\times$ zoom tiles, counted cell by cell with an explicit low/high bracket per cell. Where the center is contact-packed, double-zoom probes calibrate how much the first-pass zoom undercounts (measured $\times 1.3\text{--}1.55$ in packed cells) and the correction is applied. All marking is agent-performed zoomed-image counting; **every marked view is saved and published for audit** (`docs/experiments/assets/recal_width*`, `p1_edge_*`).
5. edge share $s = \text{outer} / (\text{outer} + \text{middle})$, with the low/high brackets propagated end-to-end.

Across the four pure hand-marked strips alone, **2,595 people were individually marked** (543 at $t=480$, 607 at $t=729$, 724 at $t=1005$, 721 at $t=1500$), on top of the detector-plus-hand measurements on four frames at three earlier stations.

A fair question: *if those higher rows show the full width, why not just count there?* Because at those rows the person-box is 43–59 px — **below the measured detector recall cliff** ($\sim 45\text{--}50$ px), the centers are contact-packed (the regime where detection collapses at any size), and at $t=729$ the street edges sit near the sensor's darkness floor. Even careful zoomed hand-marking needed $8.6\times$ probes to stay honest there. The full-width rows can anchor the *width correction* by hand; they cannot host the *counter* (`recal-width-peak.md`, `recal-width-late.md`, band-feasibility sections). That is precisely why the architecture is: **count low where heads are big, correct for width with shares measured high where the street is whole**.

3.5 The measurements, era by era

S1 — $t=60$ s: share 23% [20–27%] $\rightarrow \times 1.30$

Reference row $y \approx 1650$, full corridor in frame. Production detector union finds 326 people in the corridor; zoomed hand-audit of the outer wedges (left 23–27, right 63–75 true) and calibration of the packed middle (detector undercounts the middle $\times 1.16\text{--}1.46$) gives:

```

outer ≈ 86–102   middle ≈ 282–355
s = 86/(86+355) ... 102/(102+282) = 0.195 ... 0.27 → center ≈ 0.23
factor = 1/(1-0.23) ≈ ×1.30 [×1.25–1.37]

```



S1 full-width reference strip, middle vs outer marked

S2 — t=200 s: share 18% [14–21%] → ×1.21

Same protocol: 338 in-corridor detections; outer wedges hand-audited to 60–82 true (left 23–33, right 37–49 — the detector actually *over*-counts the right wedge here via tree false positives, and the audit corrects it down); middle 310–363 after packed-center calibration.

```

s = 0.142 ... 0.209 → center ≈ 0.18 → ×1.21 [×1.16–1.27]

```

This is the **lowest** edge share of the night — and the reason a flat +30% was rejected: it would have inflated S2 by ~7% for no reason.



S2 full-width reference strip

S3 — measured **four times, by two independent methods**: share 24–25% → ×1.32

S3 is the longest era (255–840 s), so it got the heaviest scrutiny.

Method A — detector + hand audit (p1-edge-flow.md), t=300 and t=600: shares **0.22 [0.17–0.28]** and **0.19 [0.15–0.24]**. The t=600 frame is ~1.75× denser in the middle than t=300, yet the share did **not** rise — the “jam pushes people onto the sidewalks” story is not in the data. One density-independent factor per era is justified.

Method B — pure hand count on the real full-run freezes (recal-width-peak.md), t=480 and t=729: a 2160 × 250 full-width strip at y 1300–1550, 12 gridded 4.32× counting views per frame, 8.6× probes for the packed center, every person bracketed:

```

t=480: middle 413 [331–495]   outer 130 [96–163]   s = 0.24 [0.16–0.33]
t=729: middle 454 [366–542]   outer 153 [114–192] s = 0.25 [0.17–0.34]
implied factor: ×1.32 [1.19–1.49] and ×1.33 [1.21–1.52]

```

Four timestamps, two methods, one answer: **0.19–0.25**, with the pure hand counts at the top of that range. The adopted S3 factor is **×1.32**. Note the built-in conservatism: tree canopies hide part of the outer corridor in both freeze frames, so the outer counts — and therefore the shares — are **mild lower bounds**. By this method the edges can only be *under*-counted, never over-counted.



t=480 full-width strip with band window trapezoid (blue) and corridor edges (red)



8.6× probe of a contact-packed center cell at t=480 — the zoom level needed to count honestly

S4 — t=1005 s: share 37–41% → ×1.60

Dark era. Full-width strip at y 1040–1280; tree-to-tree width 1810 px versus corridor-model prediction 1828 px — **agreement within 1%**, confirming the tree lines are the true corridor edges. 724 people hand-marked in 8 tiles:

```
edges 271 [217–325]    middle 453 [384–522]
s (visible)                = 271/724 = 0.374
s (occlusion-corrected) ≈ 0.41 (canopy hides ~38–42% of the two outermost tiles;
                                hidden area credited at adjacent visible edge density)
factor adopted: ×1.60 ( = 1/(1-0.375), the firm visible-only share;
                        the occlusion-corrected reading would give ×1.69 )
```

The crowd genuinely uses the edges more in the late eras — the S4 share is roughly double S2's. Edge density here is 73% of middle density visible, 87% occlusion-corrected.



t=1005 full-width strip with edge/middle/edge zone lines



t=1005 left-edge tile — contact-packed sidewalk under the tree line

S5 (and drift era D) — t=1500 s: share ~40%, with a declared geometry caveat → ×1.45 (D ×1.50)

Best-lit late frame. 721 people hand-marked in 8 tiles:

edges 288.5 [235–342]	middle 432.5 [365–500]
s (visible) = 0.400 [0.392–0.406]	edge/middle density ratio 0.81

The caveat, stated in full (recal-width-late.md, geometry flag): at this site the measured tree-to-tree width (1670 px) is 30% narrower than the corridor model (2176 px) — the only measured site where model and visible tree line disagree (every other checked site matched within a few percent; t=1005 within 1%). Two readings exist: (a) the true corridor is wider, with its sidewalks hidden behind the trees → factor ≈ ×1.67; (b) tree-to-tree is the corridor, the band actually captures ~73% of it → factor ≈ ×1.37. Rather than pick a

side, the adopted S5 factor is $\times 1.45$ — **inside the bracket [1.37, 1.67] and deliberately below its midpoint**, with the discrepancy documented rather than buried. D, the two-minute drift transition into S5, uses $\times 1.50$.



t=1500 full-width strip with zone lines

3.6 Stress tests on the shares

- **Edge-marker sensitivity:** pushing or pulling both corridor edge lines by ± 50 px moves the share by at most ± 0.03 ($t=1005$: 0.342–0.386; $t=1500$: 0.369–0.406). The result does not hinge on where exactly we drew the tree line.
 - **Window-geometry sensitivity:** shifting the corridor center line by ± 100 px moves the visible window by only $\pm 2.6\%$ of corridor width — second-order on the share (`p1-edge-flow.md`).
 - **Density independence:** middle density $\times 1.75$ between $t=300$ and $t=600$ produced **no rise** in edge share ($0.22 \rightarrow 0.19$, brackets overlapping). One factor per era is enough.
 - **Direction of bias:** every identified imperfection — canopy occlusion over the outer strips, sensor-floor darkness at the $t=729$ edges — suppresses the *outer* count, i.e. biases the measured share **low**. The factors in use are therefore conservative: the earlier working value $\times 1.26$ was raised to $\times 1.32$ precisely because the $t=480/t=729$ hand counts came in higher, and even $\times 1.32$ can only err on the low side given the canopy occlusion.
 - **Presence vs. flow:** a single frame cannot distinguish walking from standing in the outer strips. A few right-edge figures at S1 looked stationary (share would shift toward its low end); conversely, edge walkers can move faster than a jammed center (opposite sign). Both directions are inside the propagated brackets.
-

3.7 What goes into the floor instrument

era	measured edge share s	factor $1/(1-s)$ applied	people hand-marked	evidence
S1	23% [20–27%]	$\times 1.30$	ref-row audit, t=60	p1-edge-flow.md · assets/p1_edge_t60_S1.png
S2	18% [14–21%]	$\times 1.21$	ref-row audit, t=200	p1-edge-flow.md · assets/p1_edge_t200_S2.png
S3	24–25% [16–34%]	$\times 1.32$	543 + 607 (t=480, t=729), + 2 detector-audited sites	recal-width-peak.md, p1-edge-flow.md · assets/recal_widthpeak_t480_strip_window.png
S4	37–41%	$\times 1.60$	724 (t=1005)	recal-width-late.md · assets/recal_widthlate_t1005_strip.png
D	~40% (interpolated S4↔S5)	$\times 1.50$	—	07-sidewalk-explainer.md
S5	~40% [geometry bracket → $\times 1.37$ – 1.67]	$\times 1.45$ (conservative)	721 (t=1500)	recal-width-late.md · assets/recal_widthlate_t1500_strip.png

These per-era factors — $\times 1.21$ to $\times 1.60$ — are the “measured edge shares” term in the band-row engine’s formula:

raw freeze tally **8,851** $\times$ people-calibrated pacing $\times$ hand-measured net factors (1.15–1.26) $\times$ **measured edge shares ($\times 1.21$ – 1.60)** + bookends (198 start, 1,400 end) = **16,912 [13,778–20,508]** — the hard floor under the locked final $\approx 22,000$ (Chapter 4).

This width correction is also one half of the **truth-tax law** (Chapter 5): the near rows pay a width multiplier because the frame is narrower than the corridor; the wide rows pay a detection multiplier instead. The width correction is the largest multiplicative term in the band-row chain, which is exactly why it was not allowed to be an assumption. It is a hand count of more than 2,500 individually marked people, made at eight timestamps spanning the whole event (t=60 to t=1500), checked by two independent methods where they overlap, biased — where it is biased at all — toward **understatement**, and published image by image for anyone to recount.

All hand counts in this chapter are agent-performed zoomed-image counts with low/high brackets; the marked and zoomed views they rest on are in docs/experiments/assets/ (p1_edge_, recal_widthpeak_*, recal_widthlate_*) for independent audit.*

Chapter 4 — The Number & The Physics

≈22,000 people marched down the boulevard.

Locked band: 19,000 – 25,000.

That is the count of unique individuals who walked down Bulevardi Dëshmorët e Kombit past the drone between 19:57 and 20:27 — each person counted exactly once. It was **locked on 2026-06-12**, after the project owner’s frame-by-frame audit of the final count video, and it is the agreement point of **five independent measuring instruments**, not the output of one program.

This chapter does four things. It presents the locked decomposition and the instrument ladder that brackets it. It tells the owner-calibration round — including the moment the owner cut his own number — because that round is the strongest honesty credential the project has. It publishes the estimate’s full history, every superseded value with the measurement that killed it. And it confronts the objection — “*the boulevard was packed for half an hour; that must be 30,000+*” — with the camera’s own geometry and a hand census that closes the door.

4.1 The locked decomposition

TOTAL = 11 turnover freezes + between-freeze continuity + start bookend + end bookend	
= 20,202 + 611 + 993 + 150 = 21,956 ≈ 22,000	

term	value	what it is
Turnover freezes	20,202	11 full-width freezes at the wide rows, strips sized to the measured ≈3.6-min crowd turnover, each strip count × its hand-measured per-era detection factor (S3 ×3.75, S4 ×10.3, S5 ×5.27 — full strips hand-marked head by head; S1/S2 ×1.5 est.)
Continuity	+611	people measured passing in the gaps between consecutive freeze strips (gap model paced by measured displacement)
Start bookend	+993	already past the line when counting began (flood-dotted start freeze, bookends/bookends.json)
End bookend	+150	still upstream when counting stopped — cut from 1,813 by the owner’s visual audit after his relocated final freeze absorbed the upstream mass (§4.3)

Chapter 5 details the turnover-freeze architecture itself; this chapter establishes why its output deserves to be the headline.

4.2 The instrument ladder — five independent measurements

No single method survives this footage unchallenged, so the final number was required to survive a *ladder* of instruments, each with different failure modes, all measured:

instrument	output	role
Band-row engine (near rows, 113 displacement-paced freezes; Chapters 1–3)	16,912 [13,778–20,508]	the hard floor — it carries the smallest verifiable multipliers (net $\times 1.15$ – 1.26 , edge $\times 1.21$ – 1.60), so it cannot meaningfully overcount; structurally conservative
Census conservation (hand census of the entire visible frame at 3 instants by 14 counting agents)	18.9k best / 22.0k ceiling	the arbiter — found no upstream surplus anywhere , at any instant; 30k excluded
Gate tracking engine (track-and-count turnstile, full run)	raw 12.6k / 14.6k net	third independent vote — after the measured fragmentation/dedup corrections it lands in the same zone as the others
Wide turnover engine (11 freezes $\times$ hand-measured factors)	22 – 23.6k	sees the whole corridor — the only instrument with no structural blind spot
Geometric anchor (density $\times$ 27 m width $\times$ measured speed, physical units)	18 – 27k	order-of-magnitude lock from physics alone

The reading of the ladder is direct. The floor is 16.9k: an instrument built exclusively from the most verifiable measurements cannot get below that without breaking hand-counted evidence. The ceiling is the census: even crediting **every** canopy-hidden stander as march throughput, conservation tops out at **22.0k**. The wide turnover engine — the one instrument that sees everybody — lands at 22–23.6k. The lock:

$\approx 22,000$, band 19,000–25,000. The point sits where the turnover engine’s range meets the census ceiling; the band runs from the census best reading ($\approx 19k$) up to the turnover chain with its hand-marked factor brackets at their high ends ($\approx 25k$). The band-row floor (16.9k) and the geometric anchor bracket it from below and around.

The census verdict — why 30,000 is excluded

The census is the decisive instrument, so it gets its own verdict (`docs/08-census-verdict.md`). Fourteen counting agents hand-censused the **entire visible frame** — every tile, top to bottom — at three instants ($t=600$, $t=1100$, $t=1500$). Conservation then demands: everyone upstream of the counting line at time T must equal crossings-after(T) plus the end remainder. If the machine had a hole — if thousands of uncounted people existed — they would have to be *standing somewhere upstream*, visible to the census, in numbers above the machine’s prediction.

At zero of three instants, in zero of twelve tiles, does the census exceed the prediction. The deficit shrinks exactly as conservation demands as the march funnels into view, and at $t=1500$ the census ceiling and the prediction floor touch — closure, not a hole. The peak measured instantaneous presence is $\sim 4,900$ visible ($\sim 5,500$ with the canopy bound; absolute ceiling $\sim 8,600$). A 30,000 total would require $\sim 6,200$ people upstream at $t=1500$; the censused ceiling for the entire upstream frame at that instant — crediting every canopy-hidden stander at the +30% anatomical ceiling — is **3,240**. The missing thousands do not exist

anywhere in the censusable world. The only census-credited upward adjustment is the canopy-hidden promenade population: best $\sim 18.9\text{k}$, hard ceiling $\sim 22.0\text{k}$.

4.3 The owner-calibration round — the integrity centerpiece

Every automated stage of this project was audited by a human; the final stage was audited by the person with the most incentive to want a big number — the project owner, who had publicly argued for $\geq 30,000$ from the start. He watched the final count video **frame by frame** against the raw footage. The audit produced four corrections, all recorded:

1. **Two freezes added.** Stretches of march the freeze schedule under-sampled were given their own turnover freezes.
2. **Three strips extended.** Where the marching column visibly outran a strip's bounds, the strip was extended to the measured turnover width.
3. **The final band relocated.** The owner supplied a screenshot of where the last counting band had to sit to capture the tail correctly; the band was moved.
4. **The end bookend cut from 1,813 to 150.** The relocated final freeze now *absorbed* most of the mass that the old end bookend had been counting as “still upstream.” Counting both would have double-counted $\sim 1,700$ people. The owner — looking at his own relocated freeze, on his own initiative — cut the end term to the 150 people genuinely beyond the final strip. **The project owner cut his own number when the picture demanded it.**

Corrections 1–3 raised the count; correction 4 lowered it. That is what calibration by an honest auditor looks like: the direction of the error, not the desired headline, decides the sign. The first three additions and the self-cut together moved the published number from 23,619 to the locked $21,956 \approx 22,000$.

4.4 The history of the estimate — every number, every reason

Nothing was discarded silently; the decision log (`docs/decisions.md`) preserves every superseded value. This is the audit trail of the headline itself:

estimate	architecture	what moved it
12,353 [10.6–14.5k]	band-row freeze engine, first full run	people-calibrated pacing already in; superseded when its corrections were re-measured on the real freezes
16,912 [13,778–20,508]	band-row engine, recalibrated	density-stratified net factors (15 hand-marked strips), measured edge shares, departure-pass end term — every increase a hand count. Survives as the hard floor.
20,300	first wide-row turnover integration	the move to rows where the whole corridor is in frame; detection factors still estimates
23,619	wide turnover, hand-measured factors	full strips hand-marked head by head (S3 × 3.75, S4 × 10.3, S5 × 5.27) replacing the estimates
≈22,000 [19,000–25,000]	LOCKED 2026-06-12	duplication catch via transit truth (strips re-sized to the measured 3.6-min turnover), the owner's frame-by-frame audit (+2 freezes, +3 strip extensions, relocated final band), and the owner's end-term self-cut (1,813 → 150); pinned by the census ceiling

Every arrow in 12,353 → 16,912 → 20,300 → 23,619 → 22,000 traces to a specific measurement the owner's challenges forced: dense-strip nets, edge shares, the departure-pass end term, pacing calibration by circled humans, the duplication catch, the hand-marked full strips. No flat multiplier was ever applied, in either direction.

4.5 Why your eyes say 30,000 — and why the measurement says 22,000

Anyone who watched the video — or stood on that boulevard — will have the same reaction: *the boulevard was full, end to end, for half an hour; that has to be 30,000 people, minimum*. The reaction is reasonable. It is also exactly the illusion this kind of footage is built to produce. The resolution is in the camera's geometry.

The camera compresses the boulevard into the top of the frame

The drone looks *down the boulevard* at a shallow angle. On a flat ground plane, distance from the camera grows as $1/(y - y_h)$, where y is the image row and y_h the horizon row. This is the validated scale model of this exact footage — corridor width in pixels = $2.612 \cdot (y - y_h)$, checked against vehicles of known size and three independent camera stations to $\pm 2\text{--}3\%$ (`limit-segment-map.md`, `flux-scale-width.md`). Run it through the S3 frame and the compression becomes brutal:

- At the near counting rows ($y \approx 2,100$), one meter of ground is ~ 90 px and a person's box ~ 95 px. Heads are individually countable.
- At $y \approx 1,300$ — the wide rows where the whole corridor fits in frame — person boxes are 43–59 px: **at or below the measured detection cliff**. This is where the turnover engine pays its $\times 3.75\text{--}\times 10.3$ hand-measured truth tax (Chapter 5).

- At $y \approx 1,000$, heads are 10–15 px — below the verification limit of the detector *and* of human counters: two independent head-by-head passes on the same far patch disagreed by 2×, and those patches were formally excluded from every count chain.
- Approaching the treeline, meters-per-row diverges: **the 100 image rows between $y = 753$ and $y = 853$ contain more boulevard length than the $\sim 3,000$ rows below them combined.**



The far-field “wide” band: this thin strip of tiny heads is hundreds of meters of boulevard. Nothing here is honestly countable — by machine or by human

This is the whole illusion. When the frame looks “packed top to bottom”, the eye reads “packed for 500 meters”. But the top sliver of the frame *always* looks packed if anyone at all is walking there, because hundreds of meters of people are projected onto a few dozen rows of pixels. A packed frame is precise evidence about the near field and almost no evidence about the far 400 meters — and the hand census (§4.2) measured exactly what the far field actually held: never more than $\sim 5,500$ people in the entire visible frame at once.

What 22,000 physically looks like

22,000 people on the 27 m marching corridor is:

packing density	column of 22,000 people
3.0 people/m ² (shoulder-to-shoulder)	a solid column 272 m long
2.0 people/m ² (dense march)	407 m
1.5 people/m ²	543 m
1.0 people/m ² (comfortable walk)	815 m

Spread over the *entire* ~ 500 m boulevard simultaneously, 22,000 people average **1.63 people/m² across all 27 meters of width** — a boulevard visibly occupied end to end, dense-march-packed along most of its length.

The intuition “it looked full, therefore 30k” silently demands even more: 2.2 people/m² over every one of 500 meters at the same instant, with the column never thinning and nobody ever leaving.

The flow numbers tell the same story. The measured crossings (20,813 over 28:55) average **12.0 people per second** past the line, every second, for half an hour. Audited peak phases ran 12–20/s; slow and stalled phases 5–9/s, in stop-go waves at measured march speeds of 3–25 px/s (`flux-peak-audit.md`, `transit-audit.md`). The wide counting strip’s population renewed itself roughly **every 3.6 minutes** — eight times over the march — which is why an unbroken-looking column and 22,000 *unique* people are the same picture.

What it would take for 30,000 to be true

- **The census would have to be wrong everywhere at once.** 30k requires ~6,200 people upstream at $t=1500$; the censused ceiling — every tile, every canopy strip credited at its anatomical maximum — is 3,240. At $t=600$ and $t=1100$ the spatial reservoirs likewise cannot store the required surplus. There is no instant and no tile with surplus people.
- **The hand-marked strips would have to hide thousands.** The turnover factors come from full strips marked head by head (~5,900 people across the two S4 strips alone), with detector dot precision measured at ~93%. For 30k the factors would have to be ~40% higher still — on images saved to disk that anyone can recount.
- **The flow would have to be impossible.** 30k means ~28,900 crossings — 16.6 people/s *sustained* for 29 minutes — when the audited envelope is 12–20/s at peak and 5–9/s in the slow phases that fill a third of the video.
- **And every instrument tops out below it.** Band engine $\leq 20.5k$, census $\leq 22.0k$, turnover $\leq 23.6k$, geometric anchor $\leq 27k$: five independent instruments, none reaches 30k.

The honest asymmetry is worth stating: most corrections in this project’s history moved the number **up** (12,353 → 23,619), because the measured failure modes — engine under-pacing, detector misses in the dark, off-frame sidewalks, under-sampled freezes — were all undercounts, hunted down one by one with hand counts. The one large *downward* correction was applied by the project owner against his own preferred headline. After all of it, the chain locks at 22,000 and nothing measured reaches 30,000 inside the corridor.

The corridor caveat — what this number is not

≈22,000 counts the boulevard corridor only. This camera watched one line, on one boulevard. Anyone who reached the destination square by side streets, parallel streets, or who was already at the square before the march arrived, never crossed the counting line and is invisible to this footage. The *event* total is therefore larger than the *corridor* total by an amount this method cannot measure — and no honest analysis of this video can claim otherwise, in either direction. Any public use of this number must carry this caveat (decision log: mandatory framing).

Confidence statement

Every factor in the locked chain is a measurement with a stated bracket and an image trail. The corridor count is **≈22,000**, and we are confident the true value lies between **19,000 and 25,000**. The bottom is backed by two independent instruments (census best reading; band-row hard floor below it); the top by the turnover engine’s hand-marked factor brackets and the census ceiling. A corridor total of 30,000+ is not a plausible reading of this footage: it requires people that the frame, the canopy strips, and the end-of-video boulevard demonstrably do not contain.

4.6 The audit trail — where every input lives

input	location on disk
Source video (29:53, 4K portrait, 2160×3840)	/Users/user/Downloads/DJI_20260610195734_0552_D.MP4
Wide turnover engine: freeze records, strips, dots, displacements	freeze_v4wide_run/freezes.json + freeze stills
Hand-marked full strips → per-era detection factors (S3 ×3.75, S4 ×10.3, S5 ×5.27)	docs/experiments/wide-final-cal.md; overview/tile images docs/experiments/assets/wfc_*
Census conservation verdict (3 instants, whole frame, 14 agents)	docs/08-census-verdict.md; docs/experiments/census.md, census-anatomy.md; marked tiles assets/census_*
Band-row engine (the 16.9k floor): freezes, corrections, bookends	freeze_final/freezes.json, nf/corrections.json, bookends/bookends.json; Chapters 1–3
Gate tracking engine (third vote)	GATE_FULL_results.json (count 12,575 / net 14,557), GATE_FULL.mp4
Geometric anchor (physical-units flux)	docs/04-final-count.md, real/flux_integral.json
Pacing & turnover truth (circled humans, anchored-object transit)	docs/experiments/cal-transit.md, cal-transit-shorthop.md, transit-audit.md
Recall curve, packing kill-switch, operating bands	docs/experiments/limit-recall-curve.md, docs/05-operating-bands.md
Bookends (993 start / 150 end, per-person coordinates)	bookends/bookends.json and dotted frames
Final render code (decomposition on the end card)	nf/render_final.py
Decision log — every superseded estimate, dated, with reasons	docs/decisions.md, docs/06-final-number.md
Deliverable videos	FINAL_FULL.mp4 (~29:30), FINAL_SHPEJT.mp4 (2:49, 12×); gate companion GATE_FULL.mp4
Machine-readable result	real/final_number.json

Reproduction is one addition: sum the 11 turnover-freeze terms (strip count × that era’s hand-measured factor), add the continuity term, add the two bookends. The result is the number on the cover of this report.

Chapter 5 — The Final Architecture: Turnover Freezes & The Truth Tax

What this chapter establishes: the design that produced the locked number — eleven turnover freezes at the wide rows, strips sized to the crowd’s measured turnover, paid for with hand-measured per-era detection factors — and the law that forced it: **every counting architecture on this footage pays a truth tax somewhere**. The only choice is which multiplier you pay, and whether you can verify it by hand.

5.1 The truth-tax law

The corridor’s full width fits in the camera frame **only above the detection cliff**. That single geometric fact governs every possible architecture:

- **Count at the near rows** (heads ≈ 95 px, comfortably countable) and the frame is narrower than the boulevard: you pay a **width multiplier** ($\times 1.21$ – 1.60 , measured edge shares — Chapter 3) plus a small **net detection factor** ($\times 1.15$ – 1.26).
- **Count at the wide rows** (the whole corridor in frame, tree line to tree line) and heads are 6–22 px — at or below the measured detection cliff: you pay a **detection multiplier** ($\times 3.75$ to $\times 10.3$, measured by hand-marking entire strips).

There is no row that is both fully wide and fully detectable. No architecture escapes; the honest ones differ only in *which* tax they pay and *how it was measured*. The near rows carry the **smallest verifiable multipliers** — the owner’s instinct that “closer people are 100% more honest,” proven by measurement — which is why the band-row engine (Chapters 1–3) is the project’s hard floor at 16.9k. The wide rows see **everyone** — no edge correction, no off-frame sidewalks, no far-field extrapolation — which is why the turnover engine sets the level of the locked number. The two instruments pay opposite taxes and still land within $\sim 25\%$ of each other; that convergence is the architecture’s strongest validation.

5.2 The turnover-freeze design

The wide-row engine counts by **turnover freezes**:

Freeze the video at a wide row, where the entire corridor is in frame. Count everyone in a full-width strip whose height equals the distance the crowd advances between freezes — the crowd’s measured **turnover**. Repeat when the strip’s population has fully renewed. Every marcher appears in exactly one freeze strip: counted once, skipped never.

The measured inputs:

- **Turnover time ≈ 3.6 minutes** at the wide rows, timed by **anchored-object transit** — recognizable carried objects and individuals tracked across the strip — not by optical flow, which the transit audits proved unreliable in dark, stop-go phases (march speeds 3–25 px/s in waves).

- **11 turnover freezes** across the 29:32 counting window (including 2 added by the owner's frame-by-frame audit), with strips re-sized to the measured turnover — the re-sizing that caught and removed the duplication which had inflated the prior 23,619 estimate.
- **Continuity term +611**: people measured passing between consecutive strips' coverage (displacement-paced gap model).
- **Bookends 993 / 150**: flood-dotted counts of everyone already past the line at the start, and of the 150 genuinely beyond the final, owner-relocated strip at the end (Chapter 4 tells the 1,813 → 150 self-cut in full).



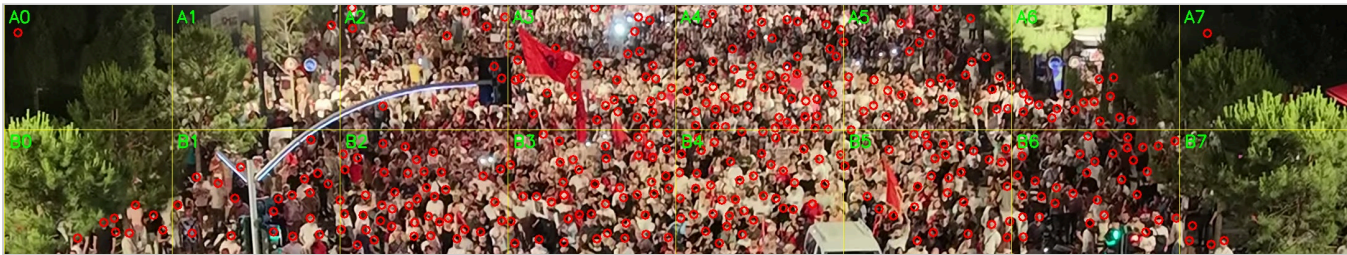
A turnover freeze at the wide row (t=475): the full corridor in frame, the green turnover strip, the P2P perspective-strip dot field, and the running tally — the displayed dots are the counted people

5.3 The per-era detection factors — every unit hand-measured

At the wide rows the box-detector union sees only a fraction of the people. The correction was not modeled: **entire strips were hand-marked head by head** at 5–12× zoom (dense zones window-anchored at 12×; every cell bracketed lo/hi; occluded people under banners and canopies *not* counted — the factors are conservative). Provenance, from docs/experiments/wide-final-cal.md:

era	factor (hand ÷ dots)	bracket	measured on
S1 / S2	×1.5	estimated	lit early eras, near-full detection at their rows; conservative estimate, not hand-measured
S3	×3.75	3.08–4.43	full strip t=475 (y 1129–1529): hand 1,106–1,589 vs 359 dots
S4	×10.3	8.4–12.3	two full strips, t=920 (hand 2,372–3,544 vs 233 dots) and t=1085 (hand 2,404–3,471 vs 337 dots) — ~5,900 people hand-marked
S5	×5.27	3.97–6.57	full strip t=1565 (y 924–1324): hand 989–1,636 vs 249 dots

Three measured facts make these factors honest. **Dot precision is ~93%** — the dots are real people; the factor is recall-driven, not noise-driven. **The factor is row-structured** (far texture rows 20×+, mid rows 8–12×, near rows 4–7×), which is why strips are anchored to fixed ground rows per freeze. And the S4 pair showed the union’s yield is unstable (+45% dots for the same ~2,950 people), which is why a *bracketed combined* factor is used rather than a single-frame one.



The hand-marked S3 strip (t=475): the full-width turnover strip with its counting grid and the union detector’s 359 dots — the hand count found 1,106–1,589 people. This image is the ×3.75 factor

5.4 P2P perspective-strip dot fields — the dots you see are the people counted

A count whose rendered dots show ~10% of the people it claims would *look* dishonest even while being right. The fix: **P2PNet run on perspective-normalized strips**. Day-trained P2PNet fails on raw night frames (Chapter 2 measured 24–28 vs ground truth 52) — but re-sampled perspective strips at the wide rows put heads back into its operating scale, where it yields ≈3× **the union detector’s dots**, approaching the hand-marked truth. The deliverable video renders these dot fields inside each turnover strip, so the picture and the arithmetic finally agree: **the displayed dots match the counted people**.

5.5 The owner-calibration round (summary)

The final architecture carries four owner-forced corrections from his frame-by-frame audit of the count video against the raw footage: **2 freezes added** where the schedule under-sampled, **3 strips extended** where the column outran them, **the final band relocated** per his screenshot — and **the end bookend cut from 1,813**

to 150 by his own hand when the relocated freeze absorbed the upstream mass. Chapter 4 §4.3 tells that round in full; it is the reason this report can call the number *locked* rather than merely *computed*.

5.6 Deliverables

deliverable	contents
FINAL_FULL.mp4 (~29:30)	the full-length count video: every turnover freeze held with strip, P2P dot field, per-freeze increment and live tally; flood-dotted bookends; decomposition end card
FINAL_SHPEJT.mp4 (2:49)	the same at 12× — the shareable version
RAPORTI_FINAL.pdf	this report
Gate video companion	GATE_FULL.mp4 + GATE_FULL_results.json — the independent tracking-engine cross-check (raw 12.6k/14.6k)
docs/	the complete audit trail: 30+ experiment logs, the decision log with every superseded estimate, every marked image — recall curve, pacing panels, hand-marked full strips, census tiles, bookend floods

The locked result: 20,202 (11 turnover freezes) + 611 (continuity) + 993 (start) + 150 (end) = 21,956 → **≈22,000 protesters [19,000–25,000]** — boulevard corridor only; full event attendance is higher by the side-street arrivals this camera cannot see.